Youyin Hu

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Education

Master of Science in Operations Research and Control Theory *Sep. 2024 – Present*
University of Shanghai for Science and Technology Shanghai, China

- GPA: 4.0/4.5
- **Relevant Courses:** Numerical Methods for Differential Equations (98/100), Statistical Learning Theory (97/100), Graph Neural Networks (93/100), Stochastic Control (91/100)

Bachelor of Science in Applied Mathematics *Sep. 2020 – Jun. 2024*
University of Shanghai for Science and Technology Shanghai, China

- **Relevant Courses:** Ordinary Differential Equations (93/100), Probability Theory (95/100), Abstract Algebra (97/100), Machine Learning (90/100), Fuzzy Mathematics (95/100)

Research Experience

A Compressed Gradient Tracking Algorithm with Differential Privacy for Distributed Optimization over Directed Graphs (*In Preparation*)
Independent Researcher *Apr. 2026 – Present*

- Extended compressed gradient tracking to directed graphs by integrating an encoding-decoding quantization scheme into the push-DIGing framework, reducing communication overhead while handling asymmetric network topologies.
- Integrating (ε, δ)-differential privacy via calibrated noise injection into push-DIGing updates, currently deriving convergence rate bounds and privacy-utility tradeoffs over time-varying directed graphs; manuscript in preparation.

Push-Sum Distributed Optimization for Multi-Agent Systems Under Encoding-Decoding Scheme
Independent Researcher *May. 2025 – Apr. 2026*

- Conceptualized and developed a novel homomorphic encryption-based push-sum framework to bridge the gap between data privacy and communication efficiency in multi-agent systems.
- Designed a unified protocol integrating Paillier encryption and encoding-decoding quantization, effectively handling asymmetric directed communication topologies.
- Derived rigorous theoretical convergence guarantees for uniformly strongly connected time-varying directed graphs, establishing explicit convergence rate bounds.
- Formulated a comprehensive error propagation analysis, proving that quantization errors decay exponentially to ensure optimal solution convergence.
- Authored the manuscript: "Push-Sum Distributed Optimization for Multi-Agent Systems Under Encoding-Decoding Scheme", submitted to *Nonlinear Dynamics* (*Under Review*).

Privacy-Preserving Distributed Recursive Filtering for State-Saturated Systems with Quantization Effects
Independent Researcher *Sep. 2024 – Feb. 2025*

- Developed a distributed recursive filter for state-saturated systems, utilizing output mask functions to ensure data privacy and scaled uniform quantization to optimize communication efficiency in sensor networks.
- Derived the upper bound of filtering error covariance via Riccati-like equations, proving its boundedness over a finite horizon and recursively solving for the optimal gain matrix to balance accuracy and stability.
- Verified the algorithm's robustness against nonlinear saturation through a three-tank system simulation in MATLAB.
- Published findings in *International Journal of Network Dynamics and Intelligence* (2025, https://doi.org/10.53941/ijndi.2025.100012).

Research Interests

- Distributed Optimization
- Privacy and Secure Computing
- Network Communication Efficiency

Publications

1. Hu, Y.; Zhang, C.; Liu, S. **Privacy-Preserving Distributed Recursive Filtering for State-Saturated Systems with Quantization Effects**. *International Journal of Network Dynamics and Intelligence* 2025, 4(2), 100012. https://doi.org/10.53941/ijndi.2025.100012
2. Hu, Y.; Liu, S.; Wang, L. **Push-Sum Distributed Optimization for Multi-Agent Systems Under Encoding-Decoding Scheme**. Submitted to *Nonlinear Dynamics* (*Under Review*)
3. Hu, Y.; Liu, S. **A Compressed Gradient Tracking Algorithm with Differential Privacy for Distributed Optimization over Directed Graphs**. (*In Preparation*)

Honors & Awards

National Third Prize, The 22nd China Graduate Mathematical Contest in Modeling 2025
China

Outstanding Dissertation, University of Shanghai for Science and Technology 2024
Shanghai, China

Teaching Experience

Teaching Assistant, Stochastic Control *Spring 2025*
University of Shanghai for Science and Technology

- Assisted Prof. Shuai Liu with course instruction for graduate-level stochastic control.
- Delivered tutorial sessions on control theory fundamentals and applications.

Skills

Programming & Analysis: Python
Mathematical Software: MATLAB, Mathematica, Maple
Scientific Writing: LaTeX
Languages: Chinese (Native), English (IELTS 7.0)